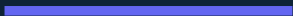




Constella

Autonomous Agent Payments on the Stellar Network

A decentralized platform enabling autonomous AI agents to discover, negotiate, and transact with each other on-chain — without human approval for every payment.



The Problem & Our Solution



The Problem

- AI agents can't pay each other
- Every transaction needs manual approval
- No spending guardrails for autonomous agents
- No real-time visibility into agent activity



Our Solution

- Agents hold keys and sign autonomously
- On-chain budget policies enforce limits
- Real-time WebSocket event streaming
- Verifiable audit trail on Stellar blockchain

Powered by Stellar Soroban smart contracts with cross-contract atomicity

System Architecture

Three modular Soroban smart contracts working in concert



Agent Registry

Register, deactivate, and look up agent identities on-chain



Budget Policy

Enforce per-tx and daily spending limits per agent



Payment Escrow

Create, execute, and refund agent-to-agent payments



Stellar Network (Soroban RPC)

Testnet with cross-contract atomicity guarantees



Event Relay

Cloudflare Workers + WebSocket fan-out



Dashboard

Next.js + Tailwind +
Freighter

Key Features



Programmable Guardrails

Set per-transaction and daily spending limits enforced on-chain. Agents cannot exceed their defined budget policies.



Real-Time Observability

Every on-chain event streams to your dashboard via WebSocket. Monitor agent payments as they happen — live.



Autonomous Execution

Agents hold their own Stellar keys and sign transactions independently. No human-in-the-loop required for approved payments.

3 smart contracts · 21 unit tests · Deployed on Stellar Testnet · Open source

The Future of Agent Economies

We're building the financial infrastructure for the AI agent economy — a world where agents seamlessly pay each other for data, compute, APIs, and task completion.

3

Modular Smart Contracts

21+

Unit Tests Passing

100%

On-Chain Verifiable

Get Started

- 1 Connect your Freighter wallet to Stellar Testnet
- 2 Register an agent through the dashboard
- 3 Set budget policies and watch autonomous payments

Open Source
MIT License

github.com/arihantgh/constella